

Assignment 5 LLM Agent (**15** Points; DDL April 20th)

CSS5120

April 2026

1 Overview

In this assignment, you will use an **LLM agent** to complete a small end-to-end NLP workflow. The purpose of this assignment is to help you transfer the ideas introduced in the lab into a computational linguistics setting. In the lab, the notebook mainly focused on how to build agent workflows and how to organize reasoning, control flow, and tool usage. In this assignment, you will apply related ideas to a text analysis problem.

A key feature of this assignment is that the **system itself should participate in data collection**. In other words, your project should not start from a fully prepared dataset given by the instructor. Instead, your LLM agent should help collect, organize, and analyze a small set of textual data on a topic of your choice. The overall task should therefore resemble a simple but complete NLP pipeline: the agent gathers data, processes it, performs one or more text analysis tasks, and then produces a final summary of findings.

You are free to design the agent workflow in your own way. You may use a single-agent design, a multi-agent design, a tool-using agent, or another reasonable LLM-agent workflow. The assignment does not require you to use a specific framework pattern. However, the final system must clearly rely on an LLM agent as a central component of the workflow.

2 Task Requirements

2.1 Choose a Base Model

You may choose any language model that you can run in your environment, such as a Hugging Face model or another accessible model suitable for Colab or local execution. You can also use the API that we bought for class for this assignment. In your report, you must clearly state the model name and briefly explain why it is appropriate for your task. Your explanation does not need to be long, but it should show that your choice is reasonable in terms of size, accessibility, language ability, or general suitability for text analysis.

2.2 Choose a Topic and Collect Text Data

You should choose a topic that can support a small NLP analysis task. For example, you may focus on opinions about AI tools, online learning, electric vehicles, a public policy issue, a product, a movie, or another topic that leads to a manageable set of short texts.

The data collection process should **involve the LLM agent in a meaningful way**. In other words, the system should not treat the dataset as a completely fixed input prepared in advance with no agent participation. Instead, the agent should play some role in the collection pipeline, such as helping retrieve texts, selecting relevant items, organizing materials from raw sources, filtering noisy content, deduplicating entries, or assisting with data cleaning.

We do not impose a strict format on how this should be done. Depending on your setup, the workflow may involve search tools, APIs, web retrieval, a crawler or script developed by LLM agent, or another reasonable mechanism for obtaining textual data. What matters is that the agent is genuinely involved in the data collection or preparation stage, rather than only being used after a fully cleaned dataset has already been prepared.

The final dataset should be text-based and should contain **at least 100 text items**. Each item should be short enough to process conveniently in your environment. You should also clearly describe

the data source and explain how the agent was involved in the collection or preparation process in your report.

2.3 Build an LLM-Agent Workflow

You should build one complete workflow in which an **LLM agent** is used to coordinate the task. The workflow should include data collection, at least one NLP analysis stage, and a final output stage. You are free to decide how the workflow is structured. For example, you may let the agent first gather relevant texts, then classify or summarize them, and finally produce an overall report. You may also let the agent call external tools or helper functions when necessary.

We do not require a particular architecture. You do not need to force your system into a multi-agent structure, and you do not need to implement a ReAct loop unless it is helpful for your design. What matters is that the system clearly functions as an **LLM-agent-based pipeline** rather than a simple one-shot prompt.

2.4 Complete at Least One NLP Analysis Task

After collecting the data, your system should perform at least **one meaningful NLP task**. Suitable tasks include sentiment analysis, topic or theme discovery, stance analysis, keyword extraction, short-text summarization, or another reasonable text analysis problem. You may perform more than one task if your workflow remains manageable, but one complete and well-executed task is sufficient.

Your analysis does not need to be highly sophisticated, and we do not assume in advance exactly how strong the agent will be on every task. Instead, this assignment is intended to give you space to explore what an LLM agent can realistically do when working with textual data. You should therefore try to go beyond simply producing raw labels whenever possible, and examine whether the system can generate results that are at least somewhat interpretable or informative.

For example, the system may identify rough sentiment patterns, recurring themes, common viewpoints, or simple differences across subsets of the collected texts. However, these are only possible directions rather than strict requirements. What matters more is that you use the assignment to explore the capabilities and limitations of the agent, and that you clearly explain what the system was able to do, what it struggled with, and how useful the output actually was.

2.5 Show the Workflow and Output

Your submission should clearly demonstrate how the system works. This includes showing the main stages of the workflow, at least one example of how the agent behaves during execution, and the final analysis result. The exact format is flexible, but your work should make it possible for the reader to understand what the agent did, what data it collected, what analysis it performed, and what conclusions it reached.

2.6 Reflection

In addition to implementation, you should briefly reflect on the strengths and limitations of your system. You may comment on whether the agent design was helpful, whether data collection was reliable, whether the model made noticeable mistakes, and how the workflow could be improved in the future. This reflection does not need to be long, but it should show some critical thinking about the design.

3 A Small Example of an LLM Agent for NLP Tasks

A simple example would be a system that studies public comments about a new AI product. In such a workflow, the agent could first retrieve a small number of public comments from a chosen source, then organize and clean them, and finally perform sentiment analysis and theme extraction. The system might end by producing a short summary describing the overall opinion distribution and the most common concerns raised by users.

This example is only intended to illustrate how the ideas from the lab may be transferred to a computational linguistics setting. You do **not** need to copy this exact design. You are encouraged to adapt the workflow to your own topic and choose a structure that makes sense for your task.

4 Report Requirements

Your report should be clearly written and should describe the task, the workflow, the data, and the results in a way that another reader can understand. At minimum, the report should explain what topic you chose, what kind of text your system collected, what model you used, and how the LLM agent was involved in the workflow.

You should also describe the final NLP task or tasks that the system performed and present the results in a readable form. A short table, a figure, or a compact summary is welcome when appropriate, but the important point is that the results should be interpretable rather than just raw model output.

In addition, the report should include at least one short example showing how the workflow actually runs. This may be an agent trace, a tool-usage example, an intermediate reasoning step, or another execution snapshot that helps the reader understand how the system behaves. Finally, the report should end with a brief discussion of what worked well, what did not work well, and how the workflow could be improved.

5 Submission

Submit a runnable notebook or Python scripts and include a **README** file with clear instructions for reproducing the results. You should also submit a PDF report following the requirements described above.

A small but complete and well-explained workflow is preferred over a complicated but unstable one. You are not expected to build a large-scale production system. What matters most is that your submission demonstrates a clear use of an **LLM agent** for collecting text data and carrying out an NLP task in a coherent workflow.